

Graph Neural Network (GNN) – Network Structure

A Graph Neural Network (GNN) is a type of neural network that works with data that has connections or relationships. Instead of looking at data like a normal table, a GNN uses a graph. A graph has different parts called nodes and edges. The nodes are the objects in the graph. The edges show the relationship or connection between the objects.

For example, if we use a social network, each person can be a node. If two people are friends, the connection between them can be an edge. A graph can look like this:

Wolf A <-> Wolf B <-> Wolf C

In this example, each wolf can represent a node, and the connections between the wolves represent the edges.

The main part of a GNN is the way information moves between connected nodes. This is called message passing. Each node receives information from its neighbors. Then, the node combines this information with its own information and updates itself.

For example, if Wolf B is connected to Wolf A and Wolf C, Wolf B can receive information from both wolves.

Wolf A > Wolf B < Wolf C

The GNN repeats this process through different layers. In the first layer, a node can learn information from its direct neighbors. With another layer, information can travel farther through the graph. A simple structure is:

Input Graph → GNN Layer 1 → GNN Layer 2 → GNN Layer 3 → Output

Each layer helps the network learn more about the graph and the relationships between the nodes. Nodes can also have their own information, called features. These features help the GNN understand what each node represents.

For example, if the nodes represent people, the features could include age, location, or other information about the person. The edges can also contain information about the relationship between two nodes. After the information passes through the GNN layers, the network can use what it learned to make a prediction.

GNNs can be used for different types of tasks, such as:

- Node classification: Predicting what type of node something is.
- Link prediction: Predicting if two nodes should have a connection.
- Graph classification: Predicting something about an entire graph.

The wolf is a good animal to represent a GNN because wolves live and work together in groups. Each wolf can be seen as a node, and the relationships between the wolves can be seen as edges. A wolf does not operate completely alone. It can learn and react based on the other wolves around it. In a similar way, a GNN learns information from the nodes that are connected to it. The basic idea of the GNN structure is:

Nodes + Edges > Message Passing > GNN Layers > Output

This is how a GNN can learn not only about individual objects, but also about the relationships between them.